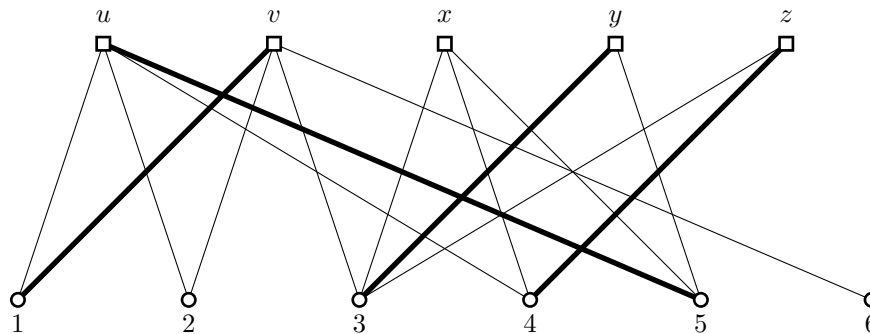


COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Solutions to worksheet for week 1.1

1. Let G be the graph shown below.



Let X be the set of vertices marked by circles, and let Y be the set of vertices marked by squares. Let M be the matching consisting of thicker edges.

Apply Algorithm 1 from the lecture notes to this input to find an augmenting path. Start by setting L to be the list $[2, 6]$ and marking every vertex as unvisited except for the vertices in L . Every time we go through the **while** loop between lines 5 and 31, we change the list L . Write down L each time you reach the end of this loop. When you process the unvisited vertices in line 9, process them in alphabetical or numerical order. What is the augmenting path produced as output by the algorithm?

Solution. The table shows the status of the list L after each repetition of the **while** loop.

Before 1st loop	$L = [2, 6]$
After 1st loop	$L = [6, u, v]$
After 2nd loop	$L = [u, v]$
After 3rd loop	$L = [v, 5]$
After 4th loop	$L = [5, 1]$

During the 5th loop, the vertex will process the unvisited vertex x , which is an unmatched vertex in Y . The algorithm terminates and returns the augmenting path $[2, u, 5, x]$.

2. Use the augmenting path you discovered in Question 1 to produce a matching that is larger by one edge. What is the matching you produce?

Solution. The matching will contain the edges $u2, v1, x5, y3, z4$.

3. Let M be a matching in a bipartite graph. Review the definition of an augmenting path. Let P be an augmenting path. Prove that if we remove from M the edges that are in P and add the edges of P that are not in M , then we obtain a matching that contains one more edge than M .

Solution. An augmenting path is a sequence of vertices where the edges that join consecutive vertices alternate between edges in the matching, and edges not in the matching. The first and last vertices in the path must not be matched.

Let M be the original matching, and let M' be the set of edges we obtain by flipping the status of edges in P .

First we show that M' is a matching. Let v be some arbitrary vertex. Remember that a matching is a set of edges where every vertex is incident with at most one of these edges. So we want to show that v is incident with at most one edge of M' .

Assume that v is not in P . Then the edges of M' incident with v are exactly the same as the edges of M incident with v . Therefore v is incident with at most one edge in M' .

Next assume that v is the first or last vertex in P . Then we only flip the status of one edge that is incident with v and this edge is not in the matching, since the first and last vertices are unmatched. Therefore v goes from being incident with no edges in M to being incident with a single edge in M' . So v is incident with exactly one edge in M' .

Next we assume that v is in the middle of P . We flip the status of the two edges in P that are incident with v . Because P is an alternating path, one of these edges is in M and the other is not. Furthermore, v is incident with exactly one edge in M , since M is a matching. So when we flip from M to M' , we remove the one edge of M that is incident with v , and we replace it with the other edge of P that is incident with v . Thus v is incident with exactly one edge in M' . We have now considered every possible case, so M' is a matching.

The fact that the first and last vertices of P are unmatched means that it has an odd number of edges and that the number of edges in P that are not in M is one more than the number of edges in P that do belong to M . This is enough to show that M' has one more edge than M .

4. The augmenting paths algorithm finds a maximum-sized matching. A matching is *maximal* if it is not possible to add another edge to it and still have a matching. Give an example of a matching that is maximal but not maximum-sized. Give an example of a graph where every maximal matching is also maximum-sized.

Solution. The matching M in Question 1 is not maximum-sized, because in Question 2 we find a matching with one more edge. However, there is no edge that we can add and still be a matching, so M is a maximal matching.

For an example of a graph where every maximal matching is also maximum sized, consider the graph with vertices $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$, where the only edges join x_1 to y_1 , x_2 to y_2 , and so on.

6. Complete truth tables for the following statements.

(a) $P_1 \Rightarrow (P_1 \wedge (\neg P_2))$

(b) $(P_1 \vee (\neg P_2)) \Rightarrow (P_2 \wedge P_3)$

(c) $(P_1 \wedge \neg(P_1)) \Rightarrow P_2$

(d) $(P_1 \wedge (P_2 \vee P_3)) \Leftrightarrow (P_1 \vee (P_2 \wedge P_3))$

Solution.

				P_1	P_2	P_3	$(P_1 \vee (\neg P_2)) \Rightarrow (P_2 \wedge P_3)$
				T	T	T	T
				T	T	F	F
				T	F	T	F
				T	F	F	F
				F	T	T	T
				F	T	F	T
				F	F	T	F
				F	F	F	F

			P_1	P_2	$P_1 \Rightarrow (P_1 \wedge (\neg P_2))$
			T	T	F
			T	F	T
			F	T	T
			F	F	T

			P_1	P_2	P_3	$(P_1 \wedge (P_2 \vee P_3)) \Leftrightarrow (P_1 \vee (P_2 \wedge P_3))$
			T	T	T	T
			T	T	F	T
			T	F	T	T
			T	F	F	F
			F	T	T	F
			F	T	F	T
			F	F	T	T
			F	F	F	T

			P_1	P_2	$(P_1 \wedge \neg(P_1)) \Rightarrow P_2$
			T	T	T
			T	F	T
			F	T	T
			F	F	T

7. If P_1 is true, P_2 is false, and P_3 is true, what is the truth value of

$$((P_1 \vee P_2) \wedge P_3) \Rightarrow (P_1 \wedge P_3) \quad ?$$

Solution. Because P_1 is true, we see that $P_1 \vee P_2$ is true. As P_3 is also true, this means that $(P_1 \vee P_2) \wedge P_3$ is true. So the antecedent of the implication is true. We Both P_1 and P_3 are true, so $P_1 \wedge P_3$ is true, meaning that the consequent is true. Since the antecedent and the consequent are both true, the implication is true.

8. Use the logical equivalences in Section 2.1 of the lecture notes to show that $\neg(P_1 \Rightarrow (P_2 \vee (\neg P_3)))$ is equivalent to $(P_1 \wedge (\neg P_2)) \wedge P_3$. At each step, state which equivalence you are using.

Solution.

$$\begin{aligned}
 \neg(P_1 \Rightarrow (P_2 \vee (\neg P_3))) &\Leftrightarrow P_1 \wedge (\neg(P_2 \vee (\neg P_3))) && \text{(Proposition 2.7)} \\
 &\Leftrightarrow P_1 \wedge ((\neg P_2) \wedge (\neg(\neg P_3))) && \text{(De Morgan's law)} \\
 &\Leftrightarrow P_1 \wedge ((\neg P_2) \wedge P_3) && \text{(Double negation)} \\
 &\Leftrightarrow (P_1 \wedge (\neg P_2)) \wedge P_3 && \text{(Associativity)}
 \end{aligned}$$

9. Use the logical equivalences in Section 2.1 of the lecture notes to show that $(P_1 \Rightarrow P_2) \vee P_3$ is equivalent to $(P_1 \wedge (\neg P_2)) \Rightarrow P_3$. At each step, state which equivalence you are using.

Solution.

$$\begin{aligned}
 (P_1 \Rightarrow P_2) \vee P_3 &\Leftrightarrow ((\neg P_1) \vee P_2) \vee P_3 && \text{(Proposition 2.7)} \\
 &\Leftrightarrow ((\neg P_1) \vee (\neg \neg P_2)) \vee P_3 && \text{(Double negation)} \\
 &\Leftrightarrow (\neg(P_1 \wedge (\neg P_2))) \vee P_3 && \text{(De Morgan's law)} \\
 &\Leftrightarrow (P_1 \wedge (\neg P_2)) \Rightarrow P_3 && \text{(Proposition 2.7)}
 \end{aligned}$$

11. (*) Let G be a bipartite graph where the vertex set is divided into two sets, X and Y , and every edge joins an edge in X to an edge in Y . Let M be a matching of G .

Give a proof that every augmenting path joins a vertex in X to a vertex in Y .

Solution. Let $v_1, v_2, v_3, \dots, v_n$ be the sequence of vertices in an augmenting path. Either v_1 is in X or it is in Y . We consider these cases separately. Assume v_1 is in X . There is an edge between v_1 and v_2 , so v_2 is in Y . There is an edge between v_2 and v_3 , so v_3 is in X . By proceeding in this way, we see that v_i is in X when i is odd, and in Y when i is even. Because the path is augmenting, we know that the edge $v_1 v_2$ is not in M , and the edge $v_2 v_3$ is in M , and so on. Therefore $v_i v_{i+1}$ is not in M when i is odd, and is in M when i is even. The final edge is not in M because the path is augmenting, so $v_{n-1} v_n$ is not in M . This means $n - 1$ is odd, and therefore n is even. Hence v_n is in Y . So in this case, the path goes from a vertex of X to a vertex of Y . If v_1 is in Y , then we could use the same arguments to show that v_n is in X . In either case, the path joins a vertex in X and a vertex in Y .

12. (*) Let M be a matching in a complete bipartite graph. Let M' be a matching that contains at least one more edge than M . Prove that there is an augmenting path for M .

(This will be proved later in the module.)

Solution. Let X be the set of edges that are in exactly one of M and M' . Any vertex is either:

- (i) incident with an edge in M , but no edge in M' ,
- (ii) incident with an edge in M' , but no edge in M ,
- (iii) incident with an edge in M and a different edge in M' ,
- (iv) incident with an edge that is in both M and M' ,
- (v) incident with no edge in M and no edge in M' .

In case (iv) or (v), the vertex is incident with no edge in X . In cases (i) and (ii), the vertex is incident with exactly one edge in X , and in case (iii), the vertex is incident with exactly two edges in X .

We think about the graph with X as its set of edges. Now every vertex in this graph is incident with zero, one, or two edges. It is not too difficult to see that any graph with this property is a collection of paths and cycles. Because M' has more edges than M , there must be a path where both the end vertices of the path are incident with edges in M' but not in M (otherwise M would have at least as many edges as M'). But this path is an augmenting path for M' .